

Photo-Synth 2

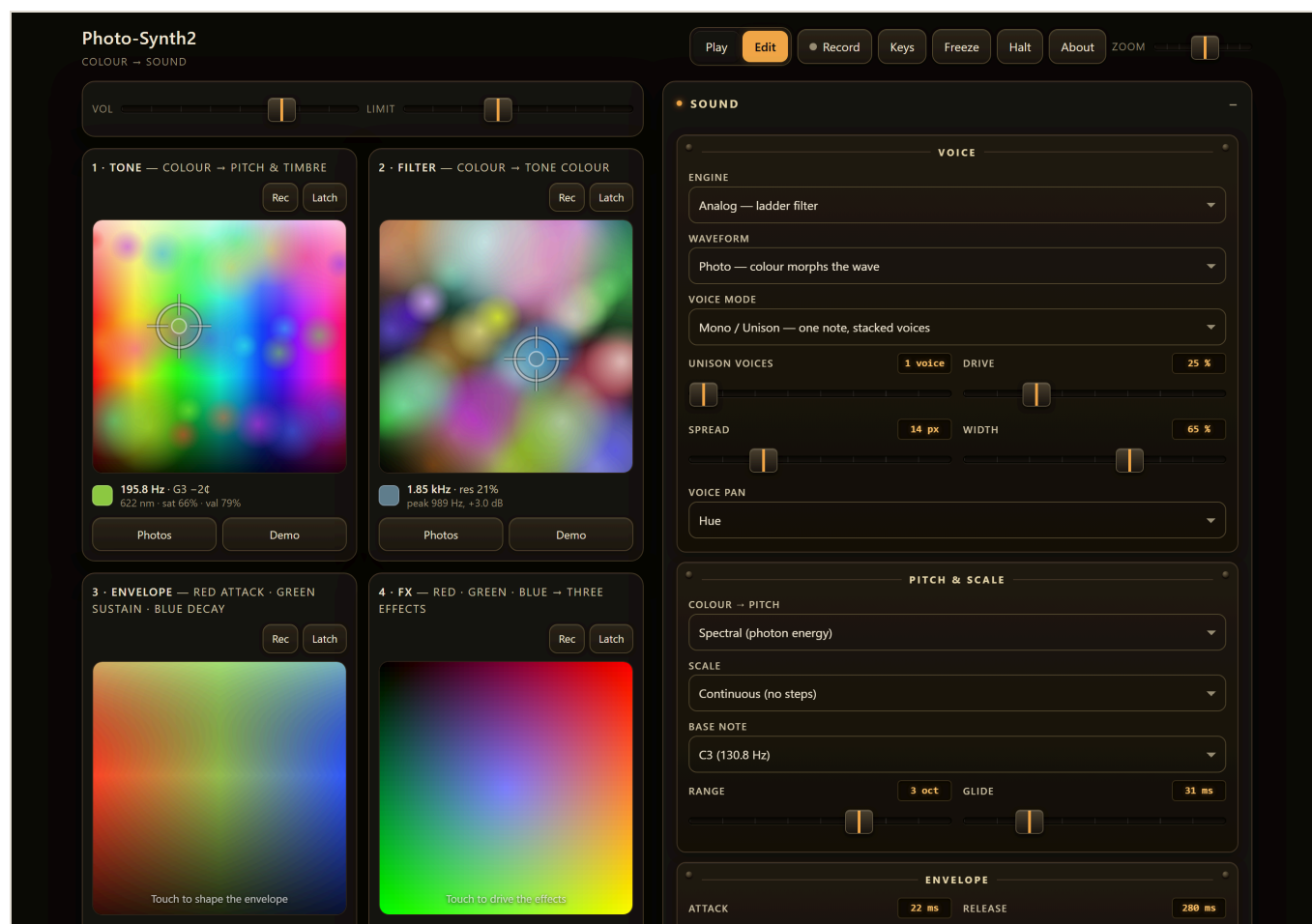
Play the colours of four photographs. A playable synthesiser that reads pitch, timbre, filter and articulation straight off a picture.

USER MANUAL

VERSION 1.0

AUGUST 2026

WINDOWS VST3



The console in Edit mode: four photo pads and the analysis strip on the left, every panel in the right-hand column. Play mode hides all of it and leaves only the photographs.

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What this instrument is

Photo-Synth 2 turns photographs into a playable synthesiser. You do not edit sound with numbers; you drag a finger — or a mouse — across a picture, and the colour under the cursor becomes the sound.

There are four photo pads, and each one listens to colour in a different way. The first is the oscillator: hue chooses the pitch, saturation morphs the waveform, brightness sets the loudness. The second is the filter. The third is the envelope. The fourth drives three effect parameters of your choosing. All four work at once, and each can be latched in place or set looping along a recorded gesture, so the instrument keeps moving on its own while you play something else.

Underneath the pictures is a complete virtual-analog voice — two band-limited oscillators that drift a few cents apart, a zero-delay-feedback ladder filter with a saturating core, a sixteen-voice pool, and a seven-module stereo effects rack you can reorder. All of it runs in native C++ inside your DAW, sample-accurate on host MIDI, and every control is automatable.

How the four pads divide the work

PAD	WHAT THE COLOUR UNDER THE CURSOR DOES
1 • Tone	Hue → pitch (red is your base note, violet the top of the range). Saturation → waveform, from a pale sine to a full saw or pulse. Green → detune and vocal edge. Brightness → loudness.
2 • Filter	Brightness → cut-off, from 80 Hz in the shadows to 14 kHz in the highlights. Saturation → resonance. Hue → a resonant peak swept from 180 Hz to 6 kHz.
3 • Envelope	Red → attack (from a 1.5 s swell down to 2 ms). Green → sustain level. Blue → decay and release (30 ms up to a 4 s tail).
4 • FX	The red, green and blue strengths each drive one effect parameter, chosen in the menus under the pad. Black is all three off, white is all three full.

Installing

Unzip the download and copy the whole **Photo-Synth2.vst3** folder into **C:\Program Files\Common Files\VST3**, then rescan plugins in your DAW. The instrument appears as **Photo-Synth2** by Brokild.

Windows 10/11, 64-bit. The editor uses the Microsoft Edge WebView2 runtime, which is already present on current Windows installations.

PRIVACY

Nothing leaves your machine. Photos are decoded and sampled locally, audio is synthesised in the plugin. There is no network access of any kind.

Quick start

Load it, drag on the first photo, and you are playing. Everything below the first two steps is refinement.

- 1 Add the instrument to a track.** It arrives with four demo photos already loaded, so it makes sound immediately — you do not need to import anything to start.
- 2 Drag slowly across photo 1.** The note follows the colour: warm reds are low, cool blues and violets are high. Lift the cursor and the note releases. This is the whole instrument in one gesture.
- 3 Rest the cursor somewhere on photo 2.** A dark patch mutes the tone, a bright patch opens it up, and a saturated colour adds resonance. Photo 2 keeps whatever colour you left it on — it is a setting, not a gesture.
- 4 Press Keys in the top bar and play.** A key sets the base note; everything the photo does is an interval above it, so the same picture can be played in any key. Chords work in Polyphonic voice mode, and MIDI from your DAW plays it too.
- 5 Switch to Edit and open Sound.** Here are the voice, the scale, the envelope and the filter type. Then open Effects for the pedal rack. Play while you change things; nothing needs to be stopped.

The top bar



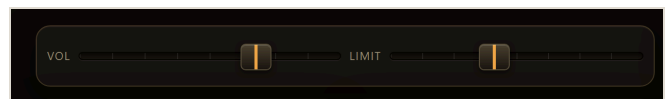
From the left: the Play / Edit switch, audio Record, the on-screen Keys, Freeze, Halt, About, and the interface Zoom.

Play / Edit	Play shows only the four photos. Edit reveals the transport, the analysis strip and all panels.
Record	Captures the plugin's own output. Save WAV in the File panel writes it as 24-bit stereo. Maximum 180 seconds.
Keys	Shows or hides the on-screen keyboard docked at the bottom.
Freeze	Holds the delay tail forever: the loop's input closes, feedback goes to unity under a soft limiter, damping opens wide. Play over the frozen bed.
Halt	A tape stop. Pitch spins down to a halt; press again to spin back up.
Zoom	Scales the whole console uniformly, for large or small plugin windows.

Volume and the limiter

Vol is the master level. **Limit** is a transparent peak limiter: pure downward gain riding above a threshold, no makeup gain and no tone shaping, so it does not colour the sound.

It is there because a resonant ladder filter with the envelope sweeping it can produce sudden peaks. At the default 40 % it catches those without ever touching ordinary playing. Turn it up if you are pushing resonance and drive hard; turn it to zero if you would rather do all the limiting yourself downstream.



Master level and limiter, at the foot of the transport.

The photo pads

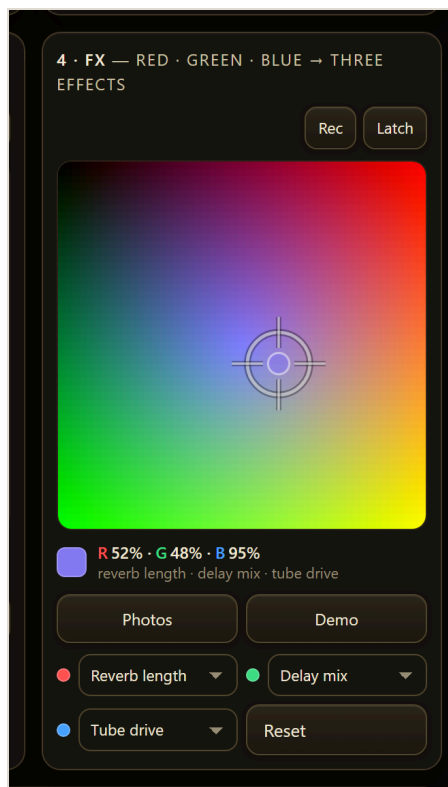
Every pad shares the same four controls — Rec, Latch, Photos and Demo — and differs only in what it does with the colour it samples.



Pad 1 · Tone. While you drag, a magnifier shows the pixels under the cursor and the readout gives the exact frequency, the nearest note with its cents offset, the sampled wavelength, saturation and brightness.



Pad 3 · Envelope. Red sets the attack, green the sustain level, blue the decay and release. Untouched, the sliders in Sound → Envelope rule instead. A red corner gives a percussive pluck, a deep blue a long ringing tail, a bright green area holds notes up indefinitely.



Pad 4 · FX. Three menus assign what red, green and blue each drive. Out of the box: red → reverb length, green → delay mix, blue → tube drive. Reset returns those sliders to their defaults and releases the pad.

Motion: gestures that loop

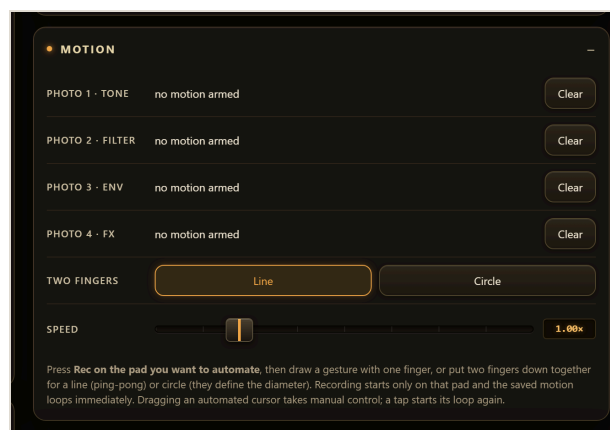
Press **Rec** on the pad you want to automate, then draw. Release, and the gesture starts looping immediately — the cursor retraces your path, sampling colour as it goes, at the speed set by Speed.

Instead of drawing you can put two fingers down together to define a straight line (which ping-pongs) or a circle (the two points set the diameter); the choice is the Two fingers switch in the Motion panel.

Every new note re-phases all running motions, so a phrase lands the same way each time. Dragging an automated cursor takes manual control without erasing the loop; a single tap starts it again. Clear discards it.

Pad controls

CONTROL	WHAT IT DOES
Photos	Loads an image from disk (JPEG, PNG, and anything else Windows can decode). The photo is stored inside the plugin, so it travels with your project and your presets.
Demo	Loads the pad's generated demo image — a hue sweep for Tone, soft colour blobs for Filter, a red-to-blue ramp for Envelope, and an RGB field for FX.
Latch	Leaves the cursor where you lifted it. On Tone the note keeps sounding; on the other pads the setting is held. Tap the pad again to move it, or switch Latch off to release.
Rec	Arms motion recording for that pad alone — see Motion, below.



The Motion panel lists what is armed on each pad, and sets the shape two fingers describe plus the global replay speed.

The analysis strip



Bars are the harmonics of the tone you are playing; the curve is the filter your second cursor is describing. Faint bars are the raw harmonic, solid bars are what survives the filter.

NOT IN THE PLUGIN

Live camera. A DAW plugin has no business opening your webcam, so the browser version's camera capture and live feed were left out. Everything else from the original is here.

The Sound panel

Five engraved units: the voice itself, how colour maps to pitch, the envelope, the Dark Drone engine, and the filter type.



The Voice unit decides what the oscillators are and how many of them there are.

Voice

Engine	Analog is the virtual-analog voice: two band-limited oscillators through a Moog-style ladder filter, running at twice the output rate. Additive is ten pure sine partials and a clean biquad — glassier, simpler.
Waveform	Photo lets the picture morph the wave from sine through triangle to saw or pulse as saturation rises. The other four — Sine, Triangle, Sawtooth, Square — pin both oscillators to that shape, band-limited and alias-free, and the photo then only affects pitch, loudness and detune.
Voice mode	Mono / Unison plays one note with stacked voices. Polyphonic plays up to four notes at once.
Unison voices	1–12. Extra voices sample the colours in rings around your cursor, so the local texture of the photo becomes a chord riding on one note. Uniform areas thicken into unison; detailed areas scatter into intervals.
Notes / Osc per note	Polyphonic mode only: how many simultaneous notes (1–4) and how many photo-sampling oscillators each note carries (1–4). Oldest note is stolen when you exceed the count.
Drive	How hard the ladder's saturating core is pushed. Thickens rather than distorts — the level is compensated.
Spread	The radius, in pixels, of the ring the extra voices sample from. Small values give a tight unison, large values a wide scatter.
Width	Stereo width. At zero the two channels are identical for true mono compatibility.
Voice pan	Whether voices are panned by the hue they sampled or by their horizontal cursor position.



Pitch & scale decides which note a given colour becomes.

Pitch & scale

Colour → pitch	Spectral reads hue as a wavelength and follows photon energy, so the intervals feel physical. Hue angle is a plain linear sweep around the colour wheel.
Scale	Continuous, Chromatic, Major, Minor pentatonic, Major pentatonic or Whole tone. Anything but Continuous snaps the colour's interval onto the scale, which makes the pads far easier to play in tune.
Base note	The note a pure red pixel plays. Also set by the keyboard, by your computer keys, and by incoming MIDI.
Range	How many octaves the full colour sweep covers, 1 to 4.
Glide	How long the pitch takes to slide when the colour changes. A little glide makes dragging sound like an instrument; none makes it sound like a scanner.



The Envelope unit. These sliders rule whenever photo 3 is not engaged.

Envelope

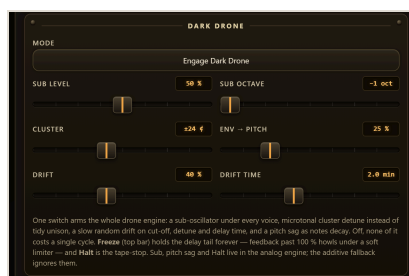
Attack / Release The basic shape, used whenever photo 3 is not engaged.

→ **Amp** How much the envelope shapes loudness. Below 100 % the level blends toward a plain click-free gate — organ-like, so the envelope can be spent on the filter instead.

→ **Filter** Sweeps the cut-off with the envelope, up to five octaves at full: closed when the envelope is down, landing exactly on your filter setting at its peak.

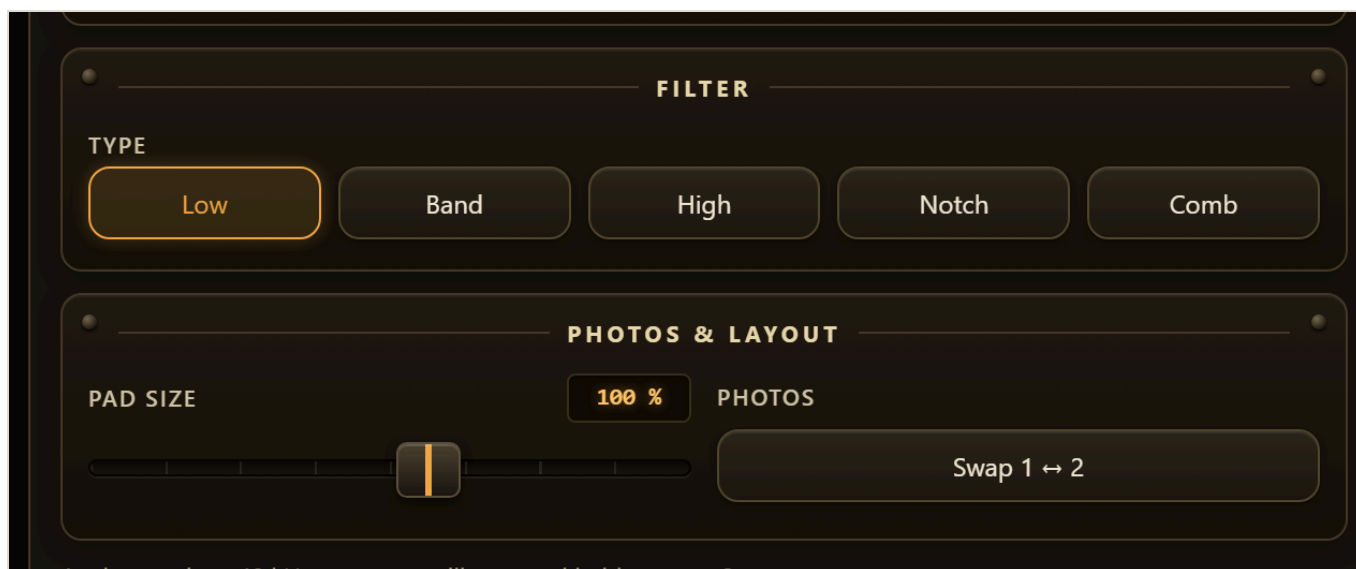
Legato Keeps the envelope alive for a moment after you lift, so tapping from colour to colour glides instead of retriggering every note.

Dark drone



One switch arms a whole drone engine, and none of it costs any CPU while it is off. **Sub level** and **Sub octave** put a thickened sine one or two octaves under every voice, through the same filter. **Cluster** replaces the tidy ± 3 -cent unison with an irregular microtonal spread of up to ± 60 cents per voice. **Env** → **Pitch** makes notes sag toward an octave down as they decay.

Drift is an ultra-slow random walk — 30 seconds to 5 minutes of characteristic time, set by Drift time — leaning on the cut-off, the detune and the delay time, so a held drone slowly comes apart on its own.



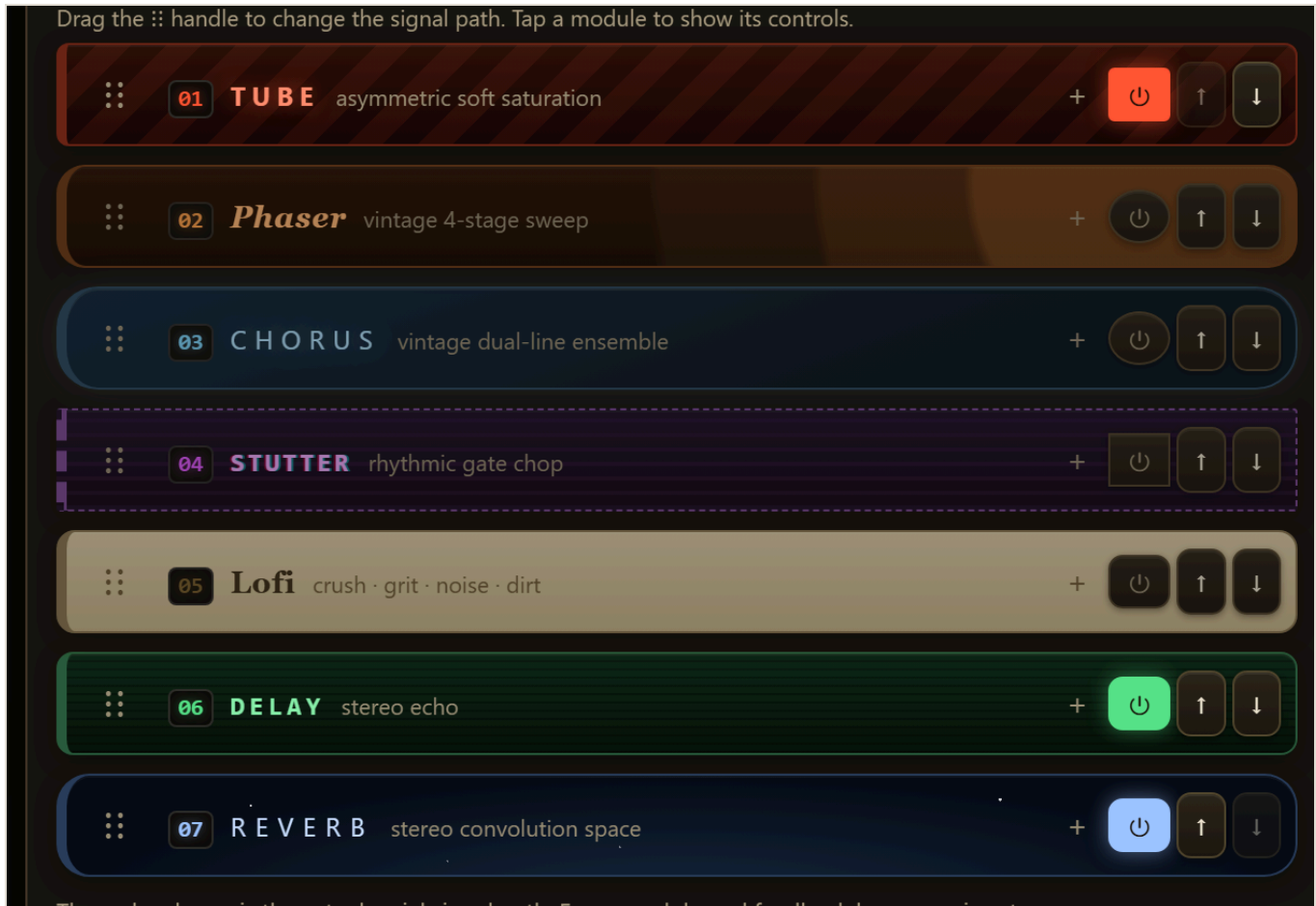
Filter type, plus Photos & layout: Pad size changes the aspect of all four pads, and Swap 1 ↔ 2 exchanges the tone and filter photos.

Filter type

Low	The classic 24 dB ladder low-pass. Dark patches of photo 2 mute the tone.
Band	Keeps a band around the cut-off and discards the rest — thin, vocal, focused.
High	Removes everything below the cut-off. Bright patches become airy and thin.
Notch	The inverse of band-pass: sweeps a hollow through the sound instead of a peak.
Comb	A tuned feedback delay whose resonances sit at every multiple of the cut-off. With high resonance it rings like a struck string — superb for metallic drones.

The effects rack

Seven stereo modules in a real serial chain. The order shown is the signal path: drag the handle, or use the arrows, and the audio graph is rewired under a short master dip so it never pops.



The rack in the Boutique skin, where every unit wears its own enclosure. Tube, Delay and Reverb start switched on; the other four start off — and a module that is off is removed from the audio graph entirely, so bypassing genuinely saves CPU. The Skin menu switches to Rack, with uniform faceplates, if you prefer the plainer look. Tap a module's name to fold its controls open or shut.

Tube — asymmetric soft saturation

Drive 0–24 dB, Tone. A biased triode-style curve adds gentle even harmonics; four-times oversampled with DC blocking. Loudness-compensated, so driving it thickens the sound without getting louder.

Delay — stereo echo

Mix, Time 40–900 ms, Feedback up to 112 %, Offset L↔R ±250 ms, Character. Past unity the loop self-oscillates into a soft limiter instead of blowing up.

Reverb — stereo convolution

Mix, Character, Length 0.2–6 s. Room, Hall, Plate, Spring (a chain of dispersive chirps — the classic boing) and Reverse, whose impulse swells in so notes arrive with a drawn-in breath.

Phaser — vintage 4-stage sweep

Mix, Rate, Depth. Four swept all-pass stages with a little feedback: the slow hollow swoosh of a seventies stomp box.

Lofi — crush · grit · noise · dirt

Crush (sample-rate reduction and bit grit), Noise (hiss plus sparse crackle), Dirt (soft normalised saturation). Everything is smoothed per sample, so the FX photo can drive it without clicks.



The Delay module, unfolded.

Chorus — vintage dual-line ensemble

Mix, Rate, Depth. Two delay lines modulated in opposite phase, wet path rolled off dark like an old bucket-brigade unit.

Stutter — rhythmic gate chop

Amount, Rate 1–16 Hz. A smoothed gate, so the chopping never clicks. Put it before delay and reverb and the tails survive the cuts.

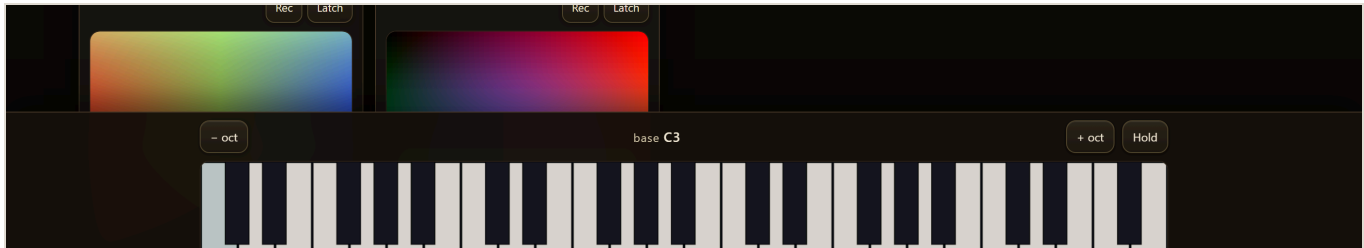
Delay character

Clean gives digital repeats. **Tape** runs the feedback loop through gentle saturation, darker damping, a low cut and a slow wow, so every repeat gets softer, dirtier and slightly unstable — a worn analog echo.

Combined with Freeze in the top bar, tape mode degenerates into a beautiful saturating murk you can play over.

Keyboard, MIDI and files

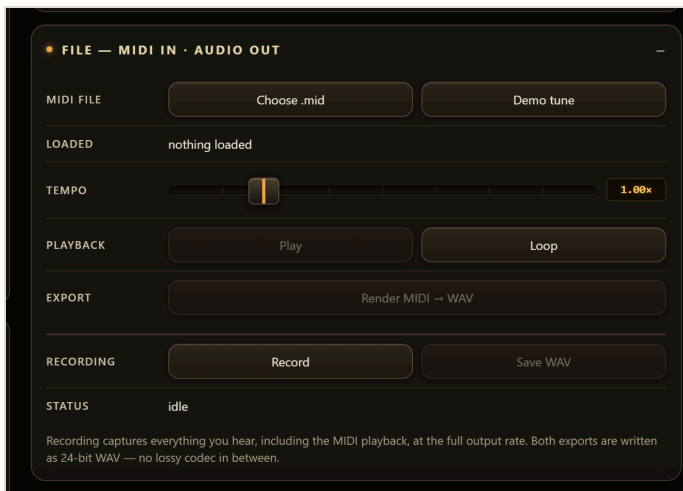
The keyboard sets the base note — the note a pure red pixel plays. It adapts to the window width, and the teal marker shows the current base.



The docked keyboard. Use – oct and + oct to move the range, and Hold to keep the last key down.

On a computer keyboard the rows `z s x d c`
`v g b h n j m` and `q 2 w 3 e r 5 t 6 y 7 u`
play two octaves from the leftmost key. In Polyphonic voice mode, several keys — or several MIDI notes — sound together as a real chord.

Your DAW plays it too. MIDI from the host is handled natively and sample-accurately, with the same note allocation as the on-screen keyboard. Record a MIDI part on the track and the photos shape every note of it.



MIDI file playback, tempo scaling, offline render and the audio recorder.

Playing a MIDI file

Choose .mid imports a standard MIDI file, or Demo tune loads a short pentatonic phrase to try the path. Its melody drives the base note while the photos keep shaping the sound; the tempo comes from the file and Tempo scales it from a quarter speed to four times. In Mono/Unison mode overlapping notes are reduced to the highest one; in Polyphonic mode the file's chords play for real, up to your Notes setting.

Recording and export

Record	Captures the plugin's output — playing, MIDI, filter sweeps and all — up to 180 seconds.
Save WAV	Writes the capture as 24-bit stereo WAV at the session's sample rate, through a normal Windows save dialog.
Render MIDI → WAV	Renders the loaded MIDI file offline, faster than real time and sample-accurate, at 48 kHz — using the colours currently under the cursors.

Presets keep everything, photographs included

A preset captures every control, the effect chain and its power switches, the FX matrix assignments, the cursor positions, the armed motions, and small copies of all four photos.

Save current stores it on this machine; **Export** writes a single `.photosynth.json` file you can back up or send to someone else; **Import file** reads one back. Your DAW project stores the same thing independently, so reopening a session restores the sound and the pictures without any preset management at all.



The Presets panel.

Try this

Nine short experiments, roughly in order of adventurousness. Each one is a handful of moves — the point is to hear what the instrument does, not to build a patch.

1 • The first thirty seconds

Drag slowly across photo 1 in a wide arc, then leave the cursor on a bright saturated area of photo 2. Now drag on photo 1 again.

Everything else in this manual is a variation on this.

2 • Play it in tune

Sound → Pitch & scale: set Scale to Minor pentatonic and Range to 2 oct. Add a little Glide.

Suddenly the photo is an instrument rather than a colour meter — every colour lands on a usable note.

3 • One photo, one chord

Voice mode Mono / Unison, Unison voices to 8, Spread to about 30 px. Put the cursor on a busy, detailed part of photo 1 — then on a smooth gradient.

Detail scatters the eight voices into a chord; smooth areas collapse them into a thick unison. The texture of the picture becomes harmony.

4 • Real chords from your DAW

Voice mode Polyphonic, Notes 4, Osc / note 2. Record a slow four-note chord progression on the track and let it play while you move the cursor on photo 1.

The chord holds while the timbre of every note travels with your cursor.

5 • A filter that plays itself

Press Rec on photo 2, put two fingers down (or click and drag with the Motion shape set to Circle) to describe a circle over a colourful area, and release. Set Speed to about 0.5x.

A slow endless filter sweep that follows the picture rather than an LFO — hold a note and listen to it breathe.

6 • Percussive from a photograph

Latch a deep red spot on photo 3 (fast attack, short tail), set Envelope → Filter to about 60 %, filter type Low, and play staccato notes on the keyboard.

Plucks and blips, with the filter snapping shut behind each note.

7 • Metallic drone

Filter type Comb. Latch photo 1 on a saturated colour and photo 2 on something bright. Engage Dark Drone, Sub level up, Cluster around 30 ¢, Drift 60 %, Drift time long. Leave it running for a few minutes.

A struck-string resonance that never quite repeats itself and slowly comes apart. This is what the instrument is best at.

8 • Freeze and play over it

Delay Character to Tape, Feedback around 90 %, Mix up. Play a phrase, then hit Freeze in the top bar and keep playing.

The phrase is held forever as a saturating tape bed while you improvise on top. Press Freeze again to let it decay.

9 • Let the fourth photo run the effects

Assign the FX pad's three menus to Lofi crush, Chorus mix and Reverb mix, switch those three modules on, then press Rec on photo 4 and draw a long wandering gesture.

The effects morph continuously along a path you drew, in combinations no three LFOs would find. Finish with Halt.

In the DAW

Automation

Around forty controls are exposed to the host as automatable parameters, named exactly as they are labelled in the panels — `vol`, `limiter`, `drive`, `attack`, `release`, `envAmp`, `envFlt`, `satGain`, `delayMix`, `delayTime`, `delayFeedback`, `reverbMix`, `reverbLength`, the phaser, chorus, stutter and lofi controls, and the Dark Drone set.

Write automation for them as you would for any synth.

WORTH KNOWING

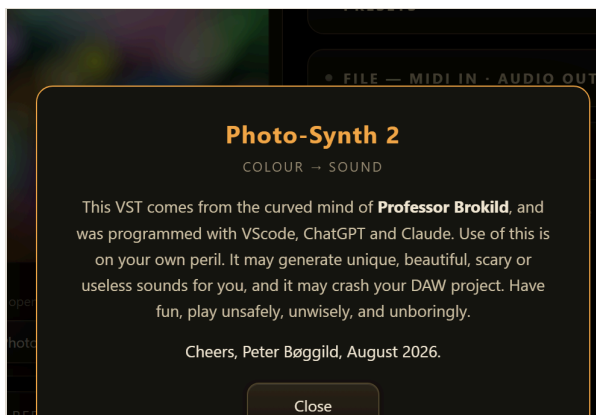
The window may be left open for gestures. The engine keeps sounding with the editor closed and host MIDI plays normally, but the parts genuinely driven by the interface — looping motions, the Dark Drone drift walk, the FX photo, and re-sampling of the photos — need the plugin window open to run. For a fixed sound, close it freely.

What the project remembers

Saving your session stores every control, the effect order and power switches, the FX matrix, the cursor positions, the armed motions and compressed copies of all four photographs. Reopening the session restores the instrument exactly, pictures included, with no external files to keep track of.

WORTH KNOWING

Resonance plus drive can peak. That is the nature of a ladder filter with the envelope sweeping it, and it is why Limit exists. It is on at 40 % by default and transparent at that setting.



The About button, in the top bar.

COLOPHON

Photo-Synth 2 · Brokild Apps · August 2026.
Native VST3 and standalone port of the browser instrument Photo Synth. The interface is the original page; the audio engine — voice pool, ladder filter, effects rack, recorder and offline render — is native C++.

This VST comes from the curved mind of Professor Brokild, and was programmed with VScode, ChatGPT and Claude. Use of this is on your own peril. It may generate unique, beautiful, scary or useless sounds for you, and it may crash your DAW project. Have fun, play unsafely, unwisely, and unboringly.

Cheers, Peter Bøggild, August 2026.